Osman Akar

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Education

PhD in Applied Mathematics at University of California, Los Angeles 2019 - Present
Research Interests: Computational linear algebra, Machine learning,
Physics/PDE based computer graphics for animations and simulations

Masters in Mathematics at University of California, Los Angeles 2016 - 2019
Departmental Scholars Program (simultaneous BS and MA degrees awarded)

BS in Mathematics of Computation at University of California, Los Angeles 2015 - 2019
Math Undergraduate Merit Scholar, Graduated Cum Laude

Projects & Research

DGCM - A Deep Gradient Correction Method for Solving Linear Systems Feb 2021 - Present
Teran Lab *Los Angeles, CA*

- **DGCM** is a research project that uses deep CNNs to accelerate iterative linear systems solvers.
- Modified classical conjugate gradient (CG) algorithm to accept machine learning input to the system.
- Designed Machine Learning Architecture and created custom loss function to perform unsupervised learning. Model learns the best search direction for the iterations in CG, which were originally chosen from krylov vectors.
- Currently **DGCM** outperforms classical CG method for linear systems with over 16 million degrees of freedom (where the linear systems comes from pressure solving part of incompressible flows in 3D in 256^3 resolution). The research is still ongoing for higher resolutions.

Research Intern at Research in Industrial Projects for Students (RIPS) Jun-Aug 2018
Institute for Pure and Applied Mathematics (IPAM) *Los Angeles, CA*

- Worked on a research team that build automated video augmentation pipeline that identifies crowd regions in sport videos and overlays an advertisement onto these regions, so that the advertisement looks aesthetic and natural for human perspective. This video link is an example of the final product. (See resulting paper arXiv:1907.09394).
- Designed and implemented an algorithm to visualize RGB-Depth image frame in 3D. Implemented *Random Sample Consensus Algorithm* in this 3D-reconstruction to detect planes to overlay advertisement onto the image in 2D.
- Developed a novel metric on video frames for the evaluation of the *PSPNet* algorithm for crowd detection.

Undergraduate Researcher at Computational and Applied Mathematics REU Jun-Aug 2017
California Research Training Program in Computational and Applied Mathematics *Los Angeles, CA*

- Worked on a research team that applied image processing techniques to extract non-content-based features from LAPD body-worn-camera video recordings and used machine learning to classify the recordings into categories.
- Modified the old machine learning model to accept the new dataset of LAPD videos. Experimented with tuning hyperparameters and reduced the error by 50%. (See the resulting paper arXiv:1904.09062)
- Created color-based feature and its metric system for video frames and implemented in the ML Algorithm.

List of Honors and Awards

- Putnam Performance Prize by UCLA Mathematics Department, 2019
- Honorable Mention at William Lowell Putnam Mathematical Competition, 2018. Ranked 77 among 4623 students.
- Richard F. Arens Putnam Scholars Undergraduate Award by UCLA Mathematics Department, 2016
- Sole recipient of UCLA Math Undergraduate Merit Scholarship in class of '19
- Gold Medal at International Mathematical Olympiad, 2014, South Africa

Programming & Technical

Languages: Python, C++, Matlab, LaTeX

Volunteer Experience

Olga Radko Math Circle (ORMC) is a free top-tier math circle attracting elementary to high school students interested in mathematics in Los Angeles Area. I am working as the Lead Instructor for the Olympiads group.